

Pod on perfboard

Same layout as the PCB, on a 20 × 43-hole perfboard: the Pico in a strip at the top (antenna clear of the screen), the screen plugged into sockets below it, the charger and DAC under the screen at the bottom edge, the battery on the back. Same Pico pins as your breadboard, so **pod.uf2 runs unchanged**.

Shopping list

Item	Qty	Notes
Perfboard, double-sided, plated holes , 2.54 mm, 9 × 15 cm	1	green FR-4, not brown paper board. Cut to 20 × 43 holes.
Female header strip 1 × 40, 2.54 mm (8.5 mm tall)	3	Pico 2 × 20, screen 1 × 14, SD 1 × 4. Each cut wastes one pin.
TP4056 charger module, USB-C, with protection	1–2	must have 6 pads: IN+, IN–, B+, B–, OUT+, OUT–
LiPo 3.7 V, 1000 mAh, size 503450 (5 × 34 × 50 mm)	1	with protection board and JST PH2.0 plug
JST PH2.0 2-pin socket with wires	1	matches the battery's plug
Slide switch SS12D00 (3 pins)	2	power switch, fitted in the case wall
Schottky diode 1N5819	2	D1
Resistor 100 kΩ, ¼ W	3	R1, R2
Ceramic capacitor 100 nF ("104")	2	soldered across R2
Wire-wrap wire, 30 AWG (Kynar), 2–3 colours	1 roll each	all signal wires on the back
Silicone wire, 24 AWG stranded, red + black	1 m each	battery, charger, switch
Solder 0.6–0.8 mm with flux, flux pen, desolder wick		
Kapton tape, double-sided foam tape, 2 mm heat-shrink		insulate the back before the battery goes on

Tools: temperature-controlled iron (330–350 °C), flush cutters, a stripper that handles 30 AWG, **multimeter**, box cutter and steel ruler.

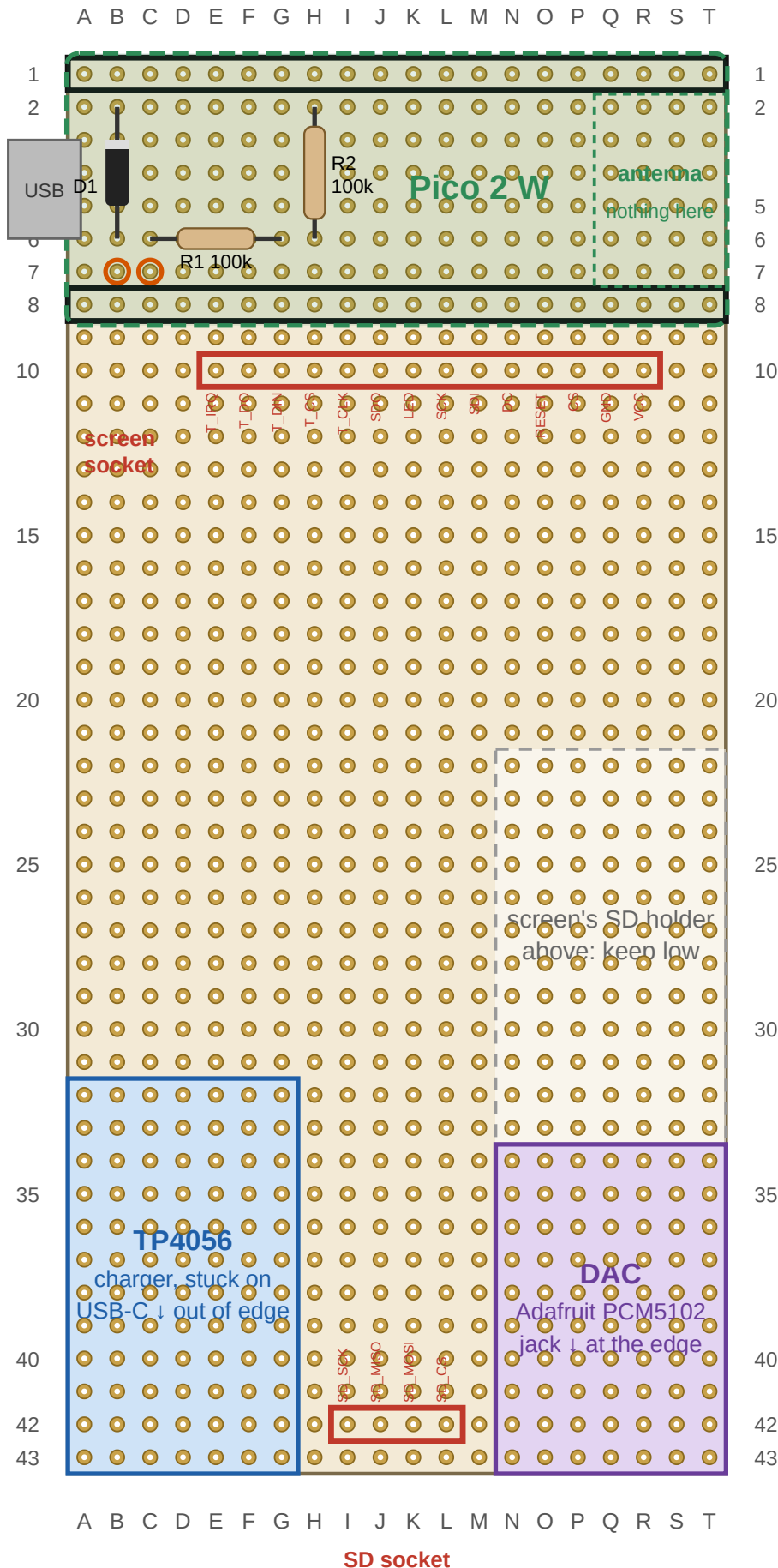
Which wire where

- **Signals (all 18 numbered wires):** 30 AWG wire-wrap wire, single core. Thin, stays where you bend it, and its insulation doesn't melt back when you solder next to it.
- **Battery, charger, switch:** 24 AWG stranded silicone. Flexible, carries over 1 A. Red for +, black for –.
- **Neighbouring holes** (marked grey on the back view): join with a blob of solder, or a scrap of cut-off resistor lead.
- No Dupont jumper wires in the final build: they're tall and work loose.

Battery and charging

- The battery only connects to the **TP4056** (B+ / B–), never straight to the Pico.
- TP4056 **OUT+** → **power switch** → **D1** → **Pico VSYS**. D1 stops the Pico's USB from pushing 5 V into the battery, so you can plug in the Pico's micro-USB to flash firmware at any time.
- Charge through the TP4056's USB-C. Red LED = charging, blue/green = full. It charges at 1 A, which is fine for a 1000 mAh cell. For a proper full charge, switch the Pod off while it charges.
- R1 and R2 halve the battery voltage so GP28 can read it. The firmware needs a small update to show the percentage from this; until then the battery icon stays hidden and everything else works.
- **Check the plug polarity** before connecting: cheap PH2.0 batteries are sometimes wired backwards. Red must reach B+. Never cut both battery wires at once.

1. Front view: where everything sits



Holes are named by column letter (A–T, left to right on the front) and row number (1–43, top to bottom). Example: **B7**.

Orange rings: B7 and C7, where the switch wire and the TP4056 OUT– wire come in.

Black boxes: two 1×20 sockets for the Pico, rows 1 and 8, all 20 columns. USB end on the left.

Red boxes: screen sockets. 1×14 in row 10, columns E–R. 1×4 in row 42, columns I–L.

Under the Pico (it sits 8.5 mm up, on its sockets): D1 from B2 (band end) to B6, R1 from C6 to G6, R2 from H2 to H6. Lay them flat.

Antenna end (right): no wires or parts on the front under it.

Bottom left: TP4056, stuck on with foam tape, USB-C poking 1 mm past the bottom edge.

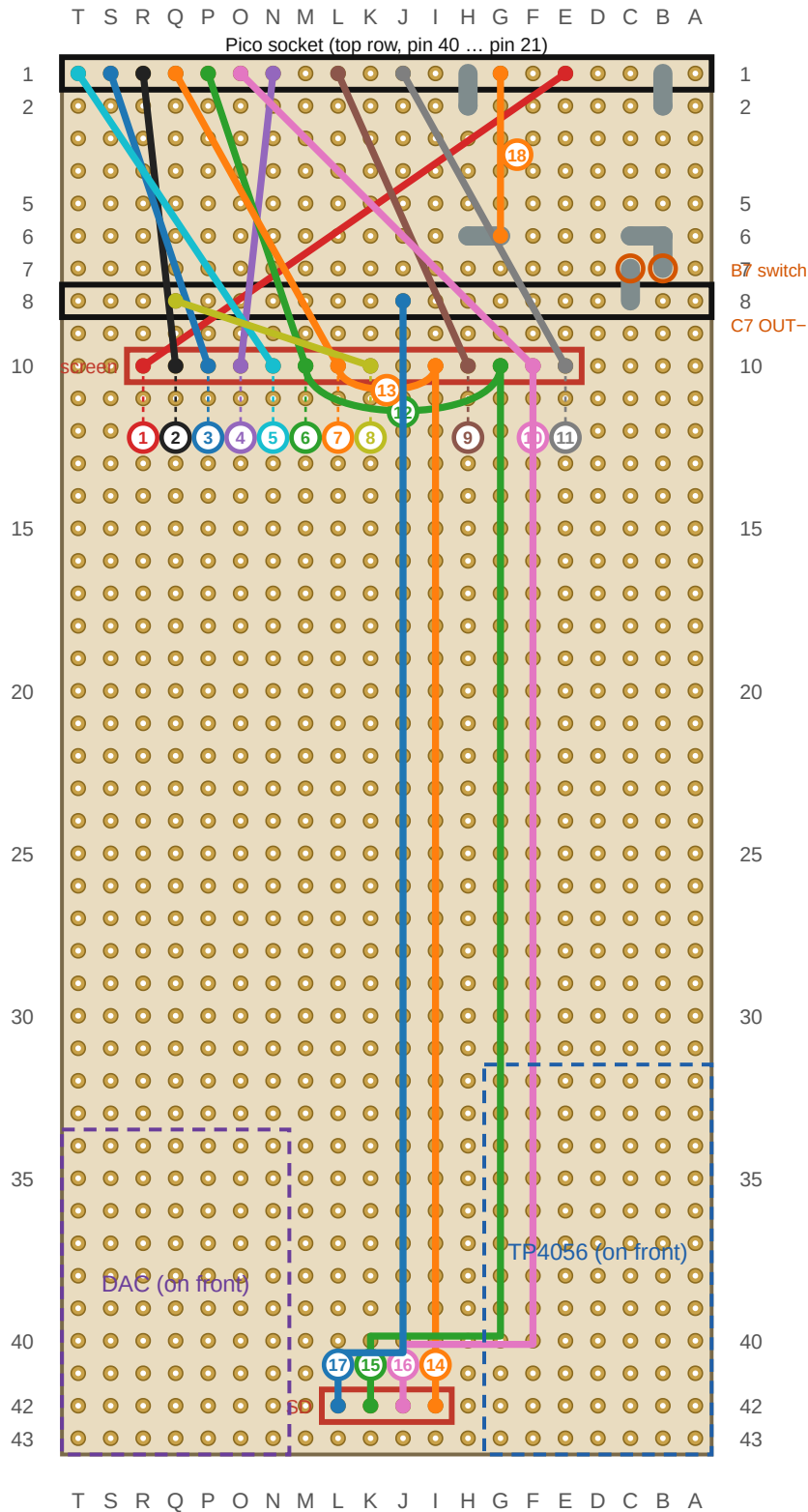
Bottom right: DAC, jack facing down at the edge.

Grey dashed area: the screen's SD-card holder sits above it. Keep it clear.

Everything under the screen must stay under **10 mm** tall.

Cutting the board: score along a line of holes on both sides, 6–8 times with the box cutter against the ruler, then snap it over a table edge and sand. You need 20 holes across and 43 down.

2. Back view: solder side (mirror image of the front)



Turn the board over: column A is now on the **right**. Every wire runs on this side.

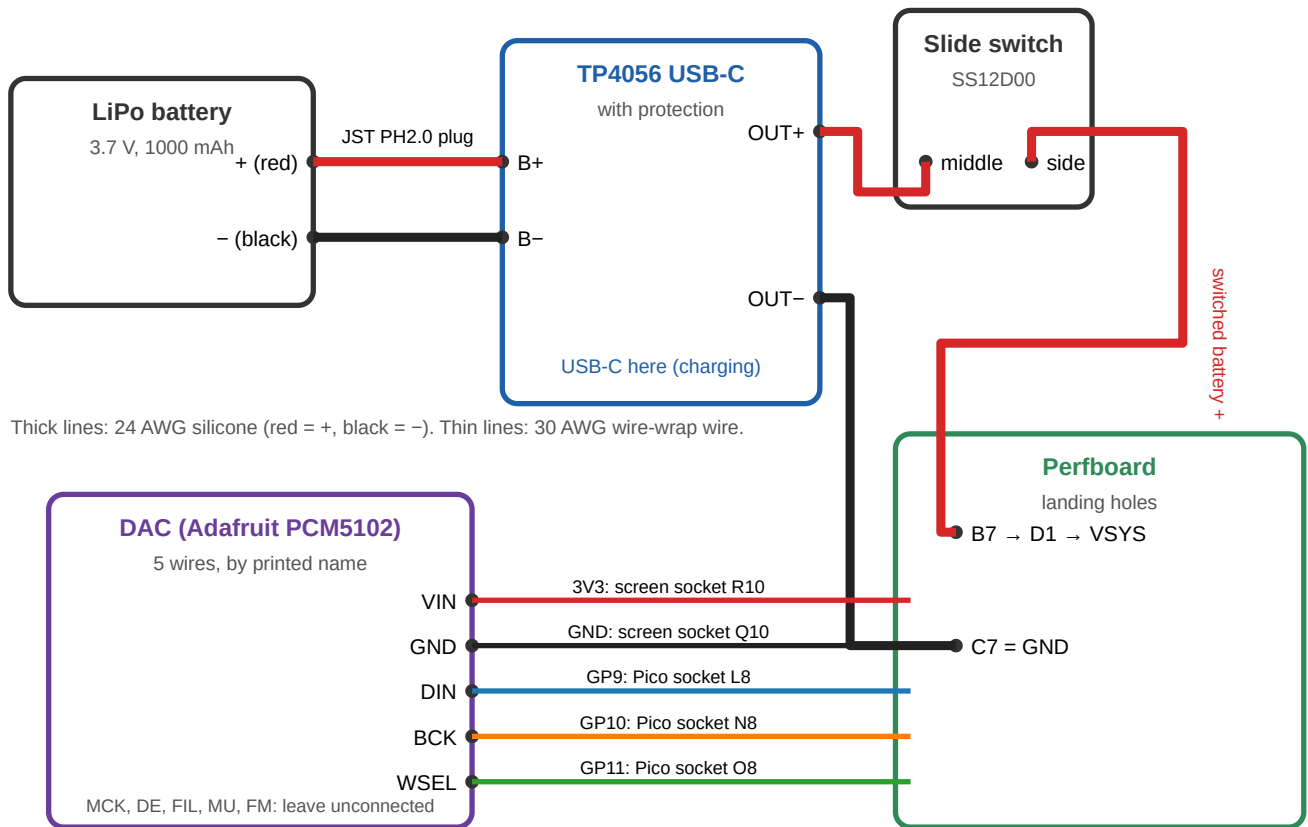
Grey bars: solder bridges between neighbouring holes: B1–B2, B6–B7, B6–C6, G6–H6, H1–H2, C7–C8.

Solder the **100 nF** capacitor straight across R2's two leads (H2 and H6) on this side.

#	From	To	Signal
1	E1	R10	3V3 → screen VCC
2	R1	Q10	GND → screen GND
3	S1	P10	GP17 → CS
4	N1	O10	GP21 → RESET
5	T1	N10	GP16 → DC
6	P1	M10	GP19 → SDI (MOSI)
7	Q1	L10	GP18 → SCK
8	Q8	K10	GP13 → LED
9	L1	H10	GP22 → T_CS
10	O1	F10	GP20 → T_DO (MISO)
11	J1	E10	GP26 → T_IRQ
12	M10	G10	SDI → T_DIN
13	L10	I10	SCK → T_CLK
14	I10	I42	T_CLK → SD_SCK
15	G10	K42	T_DIN → SD_MOSI
16	F10	J42	T_DO → SD_MISO
17	J8	L42	GP7 → SD_CS
18	G6	G1	battery sense → GP28

Ends at a socket pin: strip 2 mm, lay it against the pin, and solder. Two wires on one pin is fine.

3. Off-board wires: battery, charger, switch, DAC



Build order

1. Cut the board. Solder the two Pico sockets (plug the Pico in first so they line up, then solder).
2. Solder the 1×14 screen socket in row 10. For the 1×4 SD socket: plug it onto the screen's SD pins, plug the screen into the 1×14 socket, push the 1×4 through row 42, then solder. This lines it up exactly.
3. D1, R1, R2, the capacitor and the six solder bridges.
4. Wires 1–17. **Test:** Pico in, screen on, USB from the laptop only (no battery). Home screen, touch and SD card should work.
5. DAC and its 5 wires. **Test:** play music from the SD card on headphones.
6. Wire 18, then the TP4056, switch and battery wires. **Before plugging in the battery:** with the multimeter on continuity, check B7 to C7 does **not** beep, and check red really goes to B+.
7. Plug in the battery, switch on. Kapton tape over the back, then stick the battery on with foam tape.